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A Comparative Study on Class AB and Class D Amplifier Topologies for High Temperature Power Line Communication Circuits

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***Abstract*—Power line communication (PLC) is widely used in the well intervention industry to control tools performing maintenance on oil and gas well pipes. This paper compares PLC driver power amplifiers and the impact of temperature towards them. As both a Class AB and Class D amplifier stages form suitable candidates, a comparison is made on performance indicators, such as total harmonic distortion (THD) and intermodulation distortion (IMD), as well as power limitations and efficiency. Measurements are performed both at room temperature and at high temperature of $150\,^{\circ}\mathrm{C}$. The trade-off between Class AB and Class D based on the mentioned indicators is then further emphasized using plotted responses. These show a clear advantage in the use of a Class D amplifier topology based on efficiency by outperforming the Class AB topology by a factor of two at high temperature. However the THD is found to be approx. 25dB better in the Class AB amplifier. This article discusses and thereby identifies future research trends and topics required for high temperature PLC driving circuits.**

***Index Terms*—Class AB; Class D; High Temperature; Distortion; Efficiency**

I. Introduction

Power Line Communication (PLC) is widely used in both industry and domestic applications. PLC makes an simple connection of both power and data communication on a single conductor. The node can be powered directly and send/receive data using coupling techniques that feed this signal in to the power lines. The data signals can have various characteristics of frequencies, amplitudes and modulation strategies.

For the well intervention market robotic tools are lowered in an oil or gas well to perform maintenance at surgical precision on the pipes several kilometers below the earths surface. Lowering such a tool in to the well environment is done using a wireline, which is a heavy duty cable capable of pulling the tool back up again when the maintenance is completed. This same cable is the connection for power to, and communication with the tool. The usable bandwidth of this cable is heavily reduced by the length [1], [2]. It is found that the cable exhibits a bandiwdth up to approx. $150\,\mathrm{kHz}$, which form the baseline for this research. Due to bandwidth similarities, a comparative study can be made towards transmitter circuits that are used in the narrow band (NB-PLC).

The NB-PLC standard includes frequency ranges from $3 - 500\,\mathrm{kHz}$, which is split up in a very low frequency (VLF), low frequency (LF) and middle frequency (MF) bands [3]. The European CENELEC standard forms the closest comparison of available bandwidth formed by the wireline channel, namely $3 - 149.5\,\mathrm{kHz}$. Each of these bands may be used in various ways such as single carrier and multi carrier systems.

Wireline communication is often performed using code division multiple access (CDMA) [4] or orthogonal frequency division multiplexing (OFDM) [5], [6] where orthogonal spaced carriers are each modulated to increase the bit-rate of the connection. A low data rate (LDR) way only include a simple modulation of carriers with bit-rates up to a couple of kbps, where high data rate (HDR) connections may use complex quadrature modulation (QAM) strategies that achieve much higher bit-rates. For each of these techniques, the signal requires a highly linear output response with a high dynamic range and slew rate. The nature of any OFDM signal has a large Crest factor which is the ratio between signal peaks and average output values, also known as peak to average power ratio (PAPR).

There are many variants of PLC driver topologies that help to overcome these difficult design issues [7]. Foremost a linear amplifier, such as a Class AB amplifier, may be prefered over a switch-mode line driver solution as the output signal includes a very low amount of added harmonic artifacts. Moreover, a switch-mode line driver solution has the advantage of a higher efficiency.

Piessens et. al. [8] show an investigation of a Class D line

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driver for PLC applications. It is found that Class D amplifiers can be used for line driver applications under the criteria that either the output filter roll off is very steep (higher order filter) or that the switching frequency is considerably higher than the used bandwidth. However, higher switching frequencies may reduce the efficiency due to switching losses. The research is aimed towards an integrated solution where ambient room temperatures may be assumed.

A comparison may also be made with the audio industry, which has employed switch-mode power amplifiers for decades. The used bandwidth is slightly lower with a minimum requirement up to $20\,\mathrm{kHz}$, but measured amplifier performance is very close to what can be expected of their linear counterparts [9], [10]. Lessons can be taken from the modulation and switching investigations to optimize the performance of a PLC driver circuit.

Electronics used in well intervention systems often see a high increase in ambient temperature exceeding $150\,^{\circ}\mathrm{C}$. This forms a boundary to which extend the electronic components may be used in transmitter circuits for PLC. As can be expected, a linear amplifier is not as efficient as a switch-mode type. With an increase of temperature, losses in both types of amplifier may be expected to increase. Component usable range is heavily reduced by this extreme ambient temperature and should therefore be of high importance when designing transmitter circuits.

Works from [5] show the use of a Class A emitter follower circuit as their main driver for PLC over a 5000 ft. wireline. This type of amplifier supports the linear output requirement but it is not an efficient solution. The amplifier however shows the capability of a very high data-rate of up to 3.8Mbps. [11] shows a line driver circuit based on a Radio Frequency (RF) type linear amplifier stage equipped with a single metal oxide semiconductor field effect transistor (MOSFET), which can transmit a data rate up to 800kbps through a 7000 ft. wireline connection for logging tools.

A missing investigation of the application from switch-mode line driver circuits is addressed in this paper. A comparative study, between a linear type amplifier and a switch-mode type amplifier, is carried out. The aim of this research is to emphasize the difference and limitations of both type of amplifiers in room temperature conditions as well as high temperature conditions at $150\,^{\circ}\mathrm{C}$.

After this introduction, Section II investigates the temperature effects on both a Class AB amplifier as well as a Class D amplifier. The methods of this research are listed in Section III and the measurement results are presented in Section IV. Finally, a conclusion of the research is drafted in Section V.

II. Amplifier designs and high temperature impacts

Foremost selection criteria for an output amplifier topology are linearity, power and efficiency. Linearity is often expressed in a flat gain/phase response in the used bandwidth combined with a low THD, low IMD and low noise figures. The total output power is an amplifier design parameter, limited by the power supply to the amplifier power stage and the load. Components can be chosen accordingly to accommodate the required power levels.

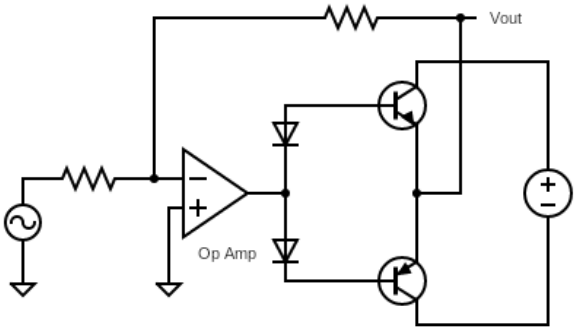


Fig. 1. An inverting Class AB amplifier stage.

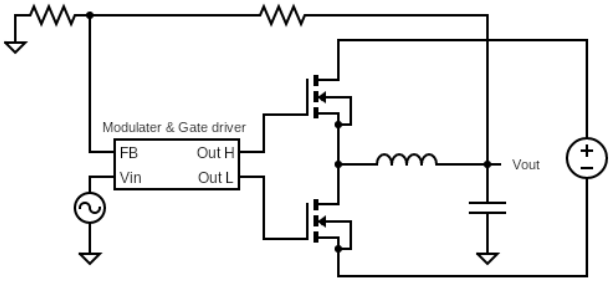


Fig. 2. An inverting Class D amplifier stage.

Efficiency is a trending topic in any type of power circuit. The need for higher efficiency circuits does not aid linearity, and is often found to be counterproductive to an amplifier's linear response. Linear amplifiers use a bias current to set the operating region of the used transistors, heavily impacting idle state losses. However, setting the bias point of output transistors in their respective linear region heavily reduces the contribution towards THD and IMD.

At high temperatures, various distortion artifacts may be introduced in the output signal. For instance thermal noise is an obvious contributor towards amplifier noise performance, but also drift capacitance over temperature is a known phenomena. Semiconductor performance in high temperature must be taken in to account when designing a high performance circuit as offset voltages may drift. For all used semiconductor components, a designer should accommodate for plenty of margin under component limits as these are also normally impacted by increasing temperatures.

A. Class AB

Class AB amplifiers, depicted in Fig. 1, have been used for decades and proved to have great linear response. A pair of bipolar junction transistors (BJTs) can be biased in their linear region to construct a clean output signal. During transition between the top and bottom transistors, a crossover

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distortion may be introduced. At high temperature, the base-emitter voltage V_{BE} will be lower than at room temperature and therefore the use of each transistor will shift. Transistors will be excited in an earlier stage during a sine wave excitation, causing a mismatch.

Due to the bias current required in the output stage, the amplifier already encounters significant losses when no output signal is present. When exciting the output stage, larger currents flow through the transistors, which are current controlled by their respective base current. Since the transistor controls a continueous current, the losses in these devices are significant. Due to the self heating of the transistors, cooling may be required. Despite the fact that for a powerful amplifier the used transistors may already have bulky packages, addition of a heatsink impacts the power density of Class AB amplifiers.

B. Class D

Class D amplifiers, depicted in Fig. 2 have been widely used in consumer and professional audio equipment. Research has shown that the performance of this type of amplifier can get close to linear Class AB type amplifier. However, Class D amplifiers significantly outperform the linear types on efficiency with the use of switch-mode techniques.

A Class D amplifier modulates the input signals with a high frequency carrier creating a pulse width modulation (PWM) signal. Because the two levels of this signal, switching transistors like field effect transistors (FET), rather than BJTs, can be used to amplify this PWM signal. The high frequency content of this carrier is filtered out at the output and the original signal waveform is then restored.

The switching stage of a Class D amplifier is equipped with FETs, rather than BJTs operating in the linear region and is widely known for its high efficiency. In idle operation, this type of amplifier can be optimized for a zero voltage switching (ZVS) regime that theoretically can reach zero Watt of idle losses. Output capacitance of each of the FETs can be charged and discharged using precision timing. In high temperature environments, these low losses are preferred, since cooling capacity is scarce.

The linearity of this type of amplifier is impacted mostly by the modulation and switching regime. A distortion in the used carrier directly correlates to an error at the output signal. The short dead-time interval that is present between the activation of the complementary switching adds to a timing error in the output. As timing is a critical parameter, parasitic elements in the switches impact distortion artifacts [12]. Passive components in the output filter inductor and capacitor cannot be expected to fully operate in the linear region as shown in [13], where temperature may impact their performance even further.

A tight control of the output is required for a adequate linear output response. A global control loop can be added to fulfil this criteria, hence making the system more complicated by adding more components.

C. Impact on communication

The distortion artifacts impact communication nodes in various ways. Amplifier clipping results in added harmonic content for higher frequency carriers where intermodulation distortion affects carriers on both sides of the fundamentals. For modulation of a carrier, in for example a QAM constellation, the vector of the modulation is altered by the distortion vector. Just like the additive noise in a channel or amplifier noise, distortion additives sum up to this vector and can therefore increase the size of a ”cloud” in a constellation graph.

III. Methods

Both type of amplifiers are built with dedicated high temperature optimized components. The supply rail voltage level is set to a total span of $30\,\mathrm{V}$. The Class AB amplifier is set up with two pairs of parallel output transistors to aid output power at high temperatures by spreading the generated losses over several devices. Due to the need of a bias current a heatsink is required. As temperature is increasing up to $150\,^{\circ}\mathrm{C}$, the heatsink is only there to minimize the ΔT between device junction and ambient temperature. The output stage is driven by an operational amplifier, which also hosts the global feedback loop. This results in a high open loop gain to lower output signal distortion.

The Class D amplifier is a high-frequency switch-mode inverter leg. The power stage consists of only two output stage transistors with minimized dimensions and a second order output filter. The switching regime is optimized for high temperature applications to minimize the losses generated in the devices. Just like the Class AB amplifier, the modulation stage is driven by a global feedback loop to boost the open loop gain of the system.

Both amplifiers are tested on bandwidth flatness, THD and IMD. Since PLC is the end goal application for these circuits, impact on a communication modulation strategy should also be taken in to account. For systems that use multiple carriers, the impact of a single non-linear element can cause problems towards other carriers in the system. Verification of the mentioned parameters is executed at both room temperature an high temperature conditions of respectively $25\,^{\circ}\mathrm{C}$ and $150\,^{\circ}\mathrm{C}$.

THD is an indicator that shows the ratio of added harmonic content to a fundamental frequency. The addition of harmonic content in a bandwidth that is filled with multiple carriers can cause problems in setting up a reliable connection. THD measured at a low carrier frequency shows added harmonic content for the higher frequencies that are still used in the band. For instance in a used bandwidth of $150\,\mathrm{kHz}$, a fundamental of $25\,\mathrm{kHz}$ THD shows up as errors for carriers that match up to the the fifth order harmonic.

In high temperature circuit design, efficiency is of high importance. Self heating can impact circuits in both life time and operational behavior. The efficiency will be estimated by measuring the input and output power of the amplifiers. The output power will be calculated from the measured voltage for each waveform amplitude in a $50\,\Omega$ load.

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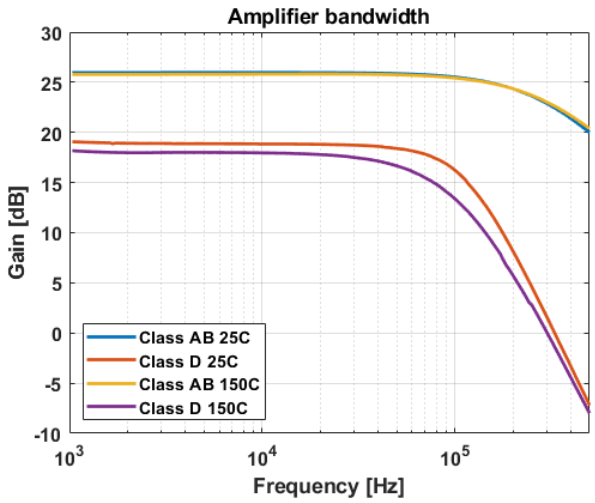


Fig. 3. Bandwidth of both amplifiers at 25 °C and 150 °C.

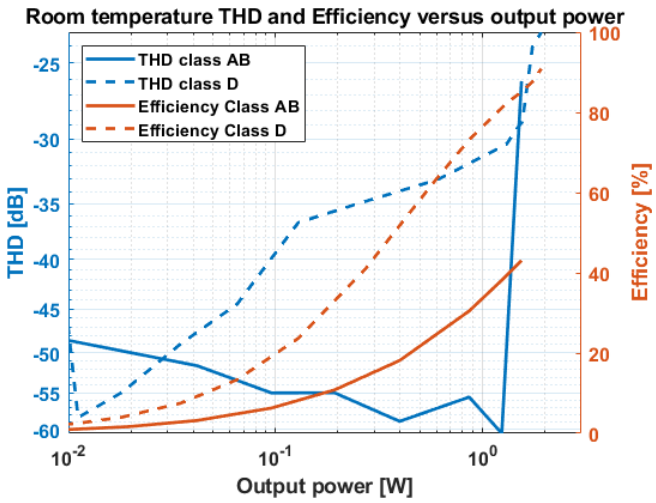


Fig. 4. THD and efficiency for Class AB and Class D amplifiers at 25 °C.

IMD summarizes the capability of a system to avoid frequency modulation due to inherent non-linear behaviour in semiconductor devices. In perspective of a multi carrier example system, two high frequency carriers (f_1 and f_2) may be transmitting. These two frequencies may create a cross-product at $2 \cdot f_1 - f_2$ and $2 \cdot f_2 - f_1$. The first cross-product may therefore end right in a lower frequency band which is used for another specific carrier. Also this contributes toward a cross error in used carrier frequencies. With $f_1 = 75\,\text{kHz}$ and $f_2 = 80\,\text{kHz}$. The measurements are processed using the Third-Order Intercept calculation.

IV. Results

The bandwidth of both amplifiers is tested at 25 °C and 150 °C. It can be observed from Fig. 3 that the gain of the Class AB amplifier is higher than the Class D, which is not a problem when investigating the output waveform based on output power. The bandwidth of the Class AB amplifier is much wider than the Class D amplifier which can be attributed to the use of a second order output filter. At high temperatures, the gain of the Class D amplifier is reduced by -2dB and bandwidth by 35 kHz.

A. Low temperature

The Class AB amplifier is more linear than the Class D counterpart. The maximum capable output level of the Class AB is lower than the Class D amplifier by 0.2 W. Output power limitation in a Class AB amplifier is formed by a higher voltage drop across the output BJT transistor stage than what is possible by turning MOSFETs fully ON. This is made visible in Fig. 4 by the steep increase in THD where the output signal shows severe clipping. The efficiency of the Class AB amplifier is significantly lower at maximum output powers by 48%, yet the THD is 30dB lower. The spectrum of the Class AB amplifier emphasizes the performance gains compared to the Class D type (Fig. 5).

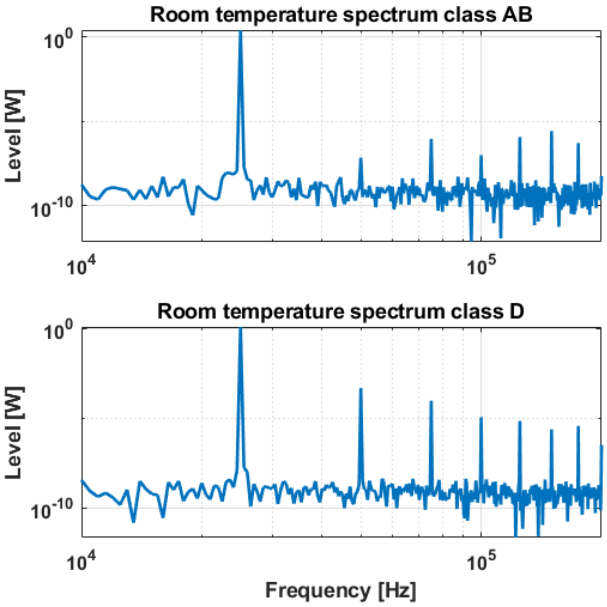


Fig. 5. Spectrum for Class AB and Class D amplifiers at 25 °C.

Finally the IMD (Third-Order Intercept) of the Class AB amplifier shows a comparable result at 28dB in Fig. 6. It must however be noted that the cross products at 10 kHz show a 5dB increase for the Class D amplifier, which also shows larger harmonic content at at 150 kHz. It is concluded that the requirements for a global control loop are much higher than for the Class AB counterpart.

B. High temperature

At high temperature both the efficiency and the maximum output power (shown in Fig. 9) is lower, respectively 5% and 0.01 W for the Class AB and 0.7% and 0.1 W for the Class D. The output transistor voltage drop at high temperature did not change significantly as the used devices are optimized for high

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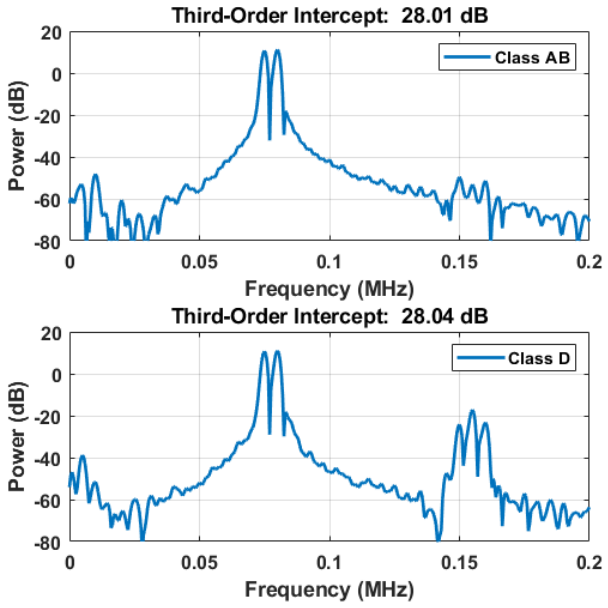


Fig. 6. IMD for Class AB and Class D amplifiers at 25 °C.

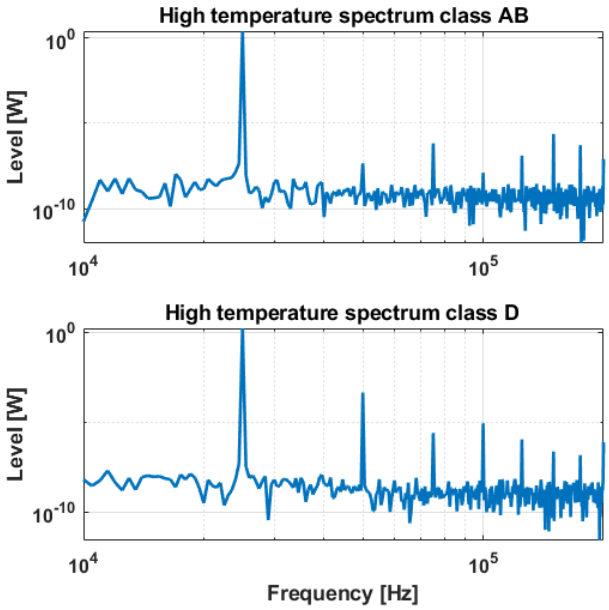


Fig. 8. Spectrum for Class AB and Class D amplifiers at 150 °C.

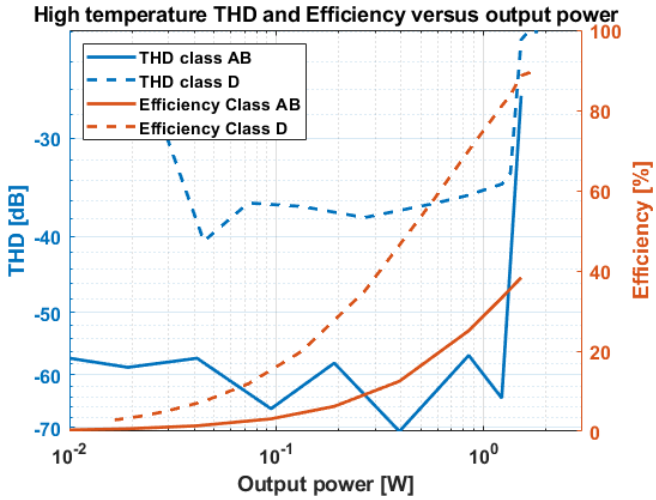


Fig. 7. THD and efficiency for Class AB and Class D amplifiers at 150 °C.

temperature operation. A decrease in output power is therefore low.

The THD levels of the Class AB amplifier in high temperature remains low, steady around the −60dB mark. The Class D amplifier distortion versus output power is increased at low output powers but the overall trend is reduced at output powers above 0.1 W. Optimization of the switching transients at high temperature proves to positively contribute to distortion reduction by 3dB at an output power of 1 W. The spectra plotted in Fig. 8 confirm this behaviour. It can be concluded the AB amplifier is more linear at these conditions but the losses are much higher.

IMD (Third-Order Intercept) of AB and Class D both see a slight reduction in high temperature. Due to limitation in the global loop compensation of the Class D amplifier, the higher added harmonic content is also decreased as compared to the room temperature measurements. Overall, the harmonic content of both amplifiers is approx. 4 to 5dB lower.

From the high temperature measurements it can be observed that the Class AB amplifier outperforms the Class D amplifier in linearity measurements. Increased losses at these temperatures increases junction temperature of devices, therefore limiting lifetime. When output power or ambient temperatures are increased further, a clear advantage of the Class D amplifier will become visible based on self-heating estimates. If linearity impairments can be improved with optimization of switching cycles and a higher loop gain in the global loop, the Class D amplifier has the potential outperforming the Class AB at extreme temperatures.

V. Conclusion

Two amplifiers have been tested in both room temperature (25 °C) and high temperature (150 °C) on their dynamic performance and power capabilities. The linear amplifier outperforms the Class D stage significantly with distortion figures up to 30dB better. However, the efficiency of the Class D amplifier is about double at a maximum of 89% in high temperature conditions. The design challenge for high temperature driver circuits for PLC applications must be focused on utilizing the power efficiency characteristics from a Class D amplifier, while improving its linear response towards the capability of a Class AB amplifier. Future studies in Class D switching cycle optimization and increase of loop gain can bring the performance of both topologies closer together. Expected losses at higher temperatures at increased output powers support that the Class D amplifier will be favorable over the Class AB. This concludes a clear trade-off between

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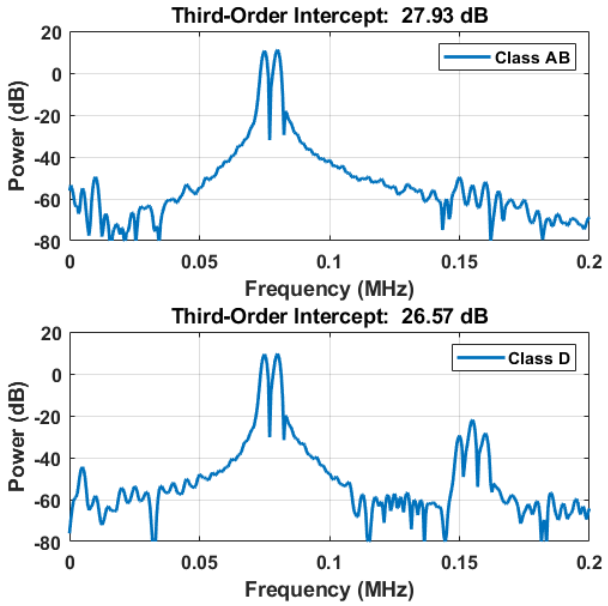


Fig. 9. IMD for Class AB and Class D amplifiers at 150 °C.

amplifier linear requirements versus power capabilities and efficiency.

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REFERENCES

[1] G. P. P. Gunarathne, E. Christidis, Y. Qureshi, A. J. Miller, and N. D. Deans, "Analysis and optimisation of transmission characteristics of cables for down-hole tools and data acquisition systems," *Imtc/99. Proceedings of the 16th Ieee Instrumentation and Measurement Technology Conference (cat. No.99ch36309)*, vol. 1, pp. 86–91 vol.1, 1999.

[2] M. S. Duraij, Y. Xiao, G. Zsurzsan, Z. Zhang, M. Hansen, and B. W. Thomsen, "A pragmatic approach to dynamic behavior in electrical lines used for well intervention tools," *Proceedings of 2020 Ieee International Conference on Industrial Technology*, vol. 2020-, pp. 601–606, 2020.

[3] G. Bucci, F. D'Innocenzo, S. Dolce, E. Fiorucci, and F. Ciancetta, "Power line communication, overview of standards and applications," *Xxi Imeko World Congress "measurement in Research and Industry"*, 2015.

[4] M.-C. Rito, R.-R. H. Joaquín, P.-M. David, and G.-G. Ricardo, "Communications System for Down-Hole Measurements," *Journal of Applied Research and Technology*, vol. 11, no. 6, pp. 903–911, 2013.

[5] A. Cashion and C. Grzegorz, "High Temperature Quadrature Amplitude Modulation over Orthogonal Frequency Division Multiplexing," 2018.

[6] R. Luo, Y. Dang, R. Li, and J. Yang, "Ofdm wireless downhole transmission systems and proposed slm method for papr reduction," *Communications and Network*, vol. 06, no. 01, pp. 9–14, 2014.

[7] E. Moons, "Looking to/for low power adsl drivers in the dslam," *Analog Circuit Design*, pp. 275–290, 2003.

[8] T. Piessens and M. Steyaert, "Class d self-oscillating line drivers," *Analog Circuit Design*, pp. 309–331, 2003.

[9] B. Putzeys, "Ncore Technology White Paper Technology White Paper," 2011.

[10] M. Duraij, N. E. Iversen, L. P. Petersen, and P. Boström, "Self-oscillating 150 w switch-mode amplifier equipped with egan-fets," *Proceedings of 139th International Audio Engineering Society (aes) Convention*, 2015.

[11] W. Zhang, Y. Shi, and J. Tang, "Application of an improved adsl technique in the high-speed wireline logging telemetry system," *2007 8th International Conference on Electronic Measurement and Instruments, Icemi*, p. 4350631, 2007.

[12] F. Koeslag, H. Du, and H. J. Beukes, "Analysis of the effect of parasitics and of reverse recovery as an amplitude error on distortion in a class D audio amplifier," *IEEE AFRICON Conference*, no. September, pp. 1–7, 2009.

[13] A. Knott, T. Stegenborg-Andersen, O. C. Thomsen, D. Bortis, J. W. Kolar, G. Pfaffinger, and M. A. E. Andersen, "Modeling distortion effects in class-d amplifier filter inductors," *128th Audio Engineering Society Convention 2010*, vol. 1, pp. 384–392, 2010.

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